Practice Assessment: Advanced SQL Queries

Scenario: You are provided with two tables: customer and orders.

- **customer** table contains: id, name.
- **orders** table contains: order_id, customer_id, product.

Task 1: Basic Retrieval & Sorting (5 Marks)

1. Write a SQL query to display all customers from the customer table, sorted by their name in **Descending** (Z to A) order.

Task 2: Joining Tables (15 Marks)

1. Write a query using an **INNER JOIN** to show the name of the customer and the product they ordered for all matching records.
2. Write a query using a **LEFT JOIN** to show all customers and any products they have ordered. This should include customers who haven't placed any orders (their product column will show NULL).
3. Write a query using a **RIGHT JOIN** to show all products ordered, including orders that might not be linked to a valid customer in the customer table.

Task 3: Grouping & Aggregation (10 Marks)

1. Using the student table structure from the slides (columns: id, name, marks), write a query to find the **Average marks** of all students and name the output column as Average_Result.
2. Write a query to **Count** how many orders each customer has placed. Display the customer_id and the count.

Task 4: Filtering Groups (10 Marks)

1. Write a query to display customer_id from the orders table only for those customers who have placed **more than 1** order. (Use GROUP BY and HAVING).

Task 5: Result Restriction (5 Marks)

1. Write a query to select only the **first 3** records from the customer table.